Vendor Websocket Protocol Document

WebSocket Stream

This section documents the real-time WebSocket stream endpoints...

Connecting to the WebSocket endpoint

```
// Javascript example.
var ws = new WebSocket("wss://<base-url>");

Headers:
Authorization = <authorization-key>

```

Base URL : internal.velora.plus

API Path : /ws/v1/vendor

Request Packet Structure

All client requests must be sent as a JSON object using the following envelope structure:

```
{
  "a": "<action>",
  "v": <values>,
  "m": "<message_id>"
}
```

Field Definitions

Field	Type	Description
`a`	string	Action to perform
`v`	array / object	Values required for the specific request type
`m`	string	Unique message identifier for this request

Action (`a`) Values

Action	Description
`subscribe`	Subscribe to one or more data streams
`unsubscribe`	Unsubscribe from one or more data streams

Message (`m`)

Message	Description
`compact_marketdata`	For compact marketdata stream

Values (`v`)

- Contains parameters required by the specific request type
- Structure depends on the message/action being sent
- Refer to the corresponding request message specification for details

Mode Values

Rule: The **first byte** of every binary message represents the `mode` and determines how the packet should be decoded and which topic it will be published to.

Mode (Byte 0)	Message Type
2	Compact Market Data

Compact Marketdata

```
Request

{ "a": "subscribe", "v": [[1, "3045"]], "m": "compact_marketdata"}

```

Bytes	Field Name	Type	JS Read Method	Description	Notes
0	msgType	int8	getInt8(0)	Message type identifier	Mode = 2 for compact marketdata
1	exchange	int8	getInt8(1)	Exchange identifier	Exchange code
2–5	instrumentToken	int32	getInt32(2)	Unique instrument token	Primary instrument ID
6–9	ltp	int32	getInt32(6)	Last traded price	Divide using price divisor if required
10–13	change	int32	getInt32(10)	Price change	Formatted value
14–17	ltt	int32	getInt32(14)	Last trade timestamp	Likely epoch time
18–21	lowDpr	int32	getInt32(18)	Day low price	lowDpr / priceDivisor
22–25	highDpr	int32	getInt32(22)	Day high price	highDpr / priceDivis

					or
26–29	currentOpenInterest	int32	getInt32(26)	Current open interest	value / qtyDivisor
30–33	initialOpenInterest	int32	getInt32(30)	Initial open interest	value / qtyDivisor
34–37	bidPrice	int32	getInt32(34)	Best bid price	value / priceDivisor
38–41	askPrice	int32	getInt32(38)	Best ask price	value / priceDivisor
42–45	ltq	int32	getInt32(42)	Last traded quantity	value / qtyDivisor
46–49	precision	int32	getInt32(46)	Precision / scaling factor	Used for formatting price/qty